Intuitively, the next theorem shows that Sinkhorn semigroups are stable as soon as the marginal-potentials (U, V) dominate at infinity the (integrated) transition-potentials (W_U, W^V). In terms of potential energy conversion, the marginal potentials should be large enough to compensate high energy transition losses. A more thorough discussion on these regularity conditions is provided in Section 4.4.

We formalize this physically intuitive principle with the second main Wasserstein-type stability theorem this article, which takes basically the following form.

Theorem 2.3. *Assume that W is locally bounded and bounded from below by a real number and (U, V) are locally bounded with compact sub-level sets. In addition, there exists some $\delta \in]0,1[$ such that the functions $(\mathcal{U}_{\delta/2}, \mathcal{V}_{\delta/2})$ are locally bounded with compact sub-level sets and we have $\lambda_{(1-\delta)U}(1) \wedge \nu_{(1-\delta)V}(1) < \infty$. Then, there exist some $\rho \in]0,1[$, $n_0 \geqslant 1$ and $c > 0$ such that for any $n \geqslant n_0$,*

$$\psi := \exp(\delta V) \Longrightarrow \|\!|\pi_{2n} - \nu_V|\!\|_\psi \leqslant c\ \rho^n \quad \text{and} \quad \|\log(d\pi_{2n}/d\nu_V)\|_\psi \leqslant c\ \rho^n. \tag{2.15}$$

When ψ satisfies the lower bound estimate (1.9) for some semi-metric ϱ then we also have the exponential Wasserstein estimates

$$\|\pi_{2n} - \nu_V\|_{tv} \vee \mathbb{W}_\varrho(\pi_{2n}, \nu_V) \leqslant c\ \rho^n. \tag{2.16}$$

In addition, the relative entropy exponential estimates (2.11) are satisfied. Similar estimates also hold for the distributions (π_{2n+1}, λ_U) with respect to the φ-norm $\|\!|\cdot|\!\|_\varphi$ associated with the function $\varphi = \exp(\delta U)$.

The estimates (2.15) are a consequence of Corollary 4.10 and the Wasserstein formulation (1.7) of weighted norms. The estimates (2.16) are a consequence of (1.10) and (2.15). More refined strict contraction estimates are established in Corollary 4.9 and Corollary 4.11.

2.4. Illustrations and comments.

Some illustrations. Section 5 provides a range of examples illustrating the stability results established in Sections 2.3 and 4. These include polynomial growth potentials (Section 5.1), heavy-tailed marginals on general normed spaces (Section 5.2), boundary state-space models (Section 5.3), semi-compact and semi-discrete settings (Section 5.7), as well as finite mixtures of statistical models (Section 5.9). Section 5.8 is devoted to linear Gaussian models, where the regularity conditions and Lyapunov criteria admit explicit characterizations.

When the transition potential W is uniformly bounded, weighted total variation norms do not provide additional insight. In this regime, exponential convergence of Sinkhorn semigroups is well understood through Hilbert's projective metric and related contraction techniques [15, 18, 46, 60], which already imply exponential decay in total variation.

On bounded subsets $\mathbb{X} \subset \mathbb{R}^d$, total variation also dominates Wasserstein distances associated with the Euclidean metric. However, the state space $\mathbb{X}$ is often non-complete and therefore non-Polish with respect to the Euclidean distance, which complicates Wasserstein convergence analysis. In this context, it is desirable to find a complete distance on the target space that makes the corresponding Wasserstein distance in distribution space complete [14].

This issue naturally motivates the introduction of alternative complete metrics. As a simple illustration, consider the Beta marginal distributions $\lambda_U = \nu_V$ on $\mathbb{X} = \mathbb{Y} =]0,1[$ equipped with the Euclidean metric $\varrho_e(x_1, x_2) = |x_1 - x_2|$ and defined by

$$\nu_V(dx) = \frac{1}{6}\ x(1-x)\ \nu(dx) \quad \text{with} \quad \nu(dx) := 1_{]0,1[}(x)\ dx.$$

Applying (2.10) to the total variation distance (see (1.14)) we readily check the Wasserstein exponential decays

$$\mathbb{W}_{\varrho_e}(\pi_{2n},\nu_V) \leqslant \|\pi_{2n}-\nu_V\|_{tv} \leqslant \rho^n \ \|\pi_0-\nu_V\|_{tv}.$$

Although the open unit interval $\mathbb{X}=]0,1[$ is not complete for the Euclidean metric, it becomes complete under

$$\varrho_c\,(x_1,x_2) := \left|\frac{1}{x_1}-\frac{1}{x_2}\right| + \left|\frac{1}{1-x_1}-\frac{1}{1-x_2}\right| \geqslant 2\ \varrho_e(x_1,x_2).$$

On the other hand, for any parameter $\delta\in]0,1[$, if we have

$$\psi(x) := \exp\left(\delta V(x)\right) \geqslant \frac{6^\delta}{2}\ \left(\frac{1}{x^\delta}+\frac{1}{(1-x)^\delta}\right)$$

then

$$\psi(x_1)+\psi(x_2) \geqslant \varrho_{c,\delta}\,(x_1,x_2) := c_\delta\ \varrho_c\,(x_1,x_2)^\delta$$

for some $c_\delta>0$. In this context, the l.h.s. estimate (2.15) yields the $\varrho_{c,\delta}$-Wasserstein estimates

$$2^\delta c_\delta\ \mathbb{W}_{\varrho_e}(\pi_{2n},\nu_V) \leqslant \mathbb{W}_{\varrho_{c,\delta}}(\pi_{2n},\nu_V) \leqslant |||\pi_{2n}-\nu_V|||_\psi \leqslant c\ \rho^n$$

as well as the weighted total variation norm decays

$$\|\pi_{2n}-\nu_V\|_{tv} \leqslant c_1\ |||\pi_{2n}-\nu_V|||_\psi \leqslant c_2\ \rho^n \quad \text{for some } c_1,c_2>0.$$

Theorem 2.3 also applies to Gaussian marginal distributions on $\mathbb{Y}=\mathbb{X}=\mathbb{R}^d$ and a transition potential W with at most quadratic growth (cf. Section 5.1). In this context, for $(p,q)=(2,1)$ in (1.11), we obtain exponential convergence in the the 2-Wasserstein distance,

$$\mathbb{W}_2(\pi_{2n},\nu_V) \leqslant c_2\ \rho_2^n \quad \text{for some parameters } c_2>0 \text{ and } \rho_2\in]0,1[.$$

Section 5.8 provides an explicit description of the functions $(\mathcal{U}_\delta,\mathcal{V}_\delta)$ in the context of linear-Gaussian models. For instance, for one-dimensional centered Gaussian marginals with unit variance and a symmetric Gaussian reference transition $\mathcal{Q}$ with variance $\tau>2$ we have

$$2/\tau<\delta<1 \Longrightarrow \mathcal{U}_{\delta/2}(x) = c_\delta + \frac{x^2}{2}\ \left(\frac{\delta}{2}-\frac{1}{\tau}\right) \stackrel{|x|\to\infty}{\longrightarrow} +\infty \quad \text{for some } c_\delta\in\mathbb{R}.$$

A symmetric Gaussian reference transition $\mathcal{Q}$ with variance τ can be interpreted as the transition probability of a Brownian motion on a time horizon τ. In this context, the exponential stability of Sinkhorn bridges w.r.t. the relative entropy for log-concave marginals at infinity, including Gaussian marginals, also holds for any time horizon $\tau>0$, see for instance [22, 26, 45]. The article [2] also provides Wasserstein exponential decays for any Gaussian marginals and any linear-Gaussian reference transition. For these classes of log-concave-type models, our Lyapunov approach yields new weighted total variation norm estimates for sufficiently large time horizons.

Energy barriers. Sections 2.3 and 5 demonstrate that Lyapunov techniques provide a flexible tool for establishing quantitative stability of Sinkhorn semigroups. Nevertheless, several important limitations remain.

Firstly, note that Theorem 2.3 applies when W is locally bounded. But even well defined absolutely continuous reference transitions on totally bounded state spaces such as $\mathbb{X}=\mathbb{Y}=]0,1[$ may have a potential W which is not well defined on the whole space. Indeed, consider the reference Markov transition defined for any $x\in]0,1[$ by

$$\mathcal{Q}(x,dy) = \frac{1}{q(x)}\ |x-y|\ \nu(dy)$$

$$\text{with} \quad \nu(dy) := 1_{]0,1[}(y)\ dy \quad \text{and} \quad \frac{1}{4}\leqslant q(x) = \frac{1}{2}-x(1-x) \leqslant \frac{1}{2}.$$

This rather elementary example is a a slight modification of (2.12) but in this case the function W is not defined on the diagonal and for any $x \neq y$ we have

$$W(x,y) = -\log|x-y| + \log q(x) \Longrightarrow W(x,x) = +\infty.$$

To the best of our knowledge, the stability analysis of Sinkhorn bridges in the context of energy-barrier cost functions of this form remains an open research question. Absolutely continuous target marginals may also have an infinite potential, for instance

$$\lambda_U(dx) := 4\ |x-1/2|\ \lambda(dx) \quad \text{with} \quad \lambda(dx) := 1_{]0,1[}(x)\ dx$$

$$\text{yields } U(x) = -\log 4 - \log|x-1/2| \Longrightarrow U(1/2) = +\infty.$$

In this case the potential function U is not well defined on the whole state space, thus this marginal model cannot be written in terms of a Boltzmann-Gibbs measure of the form (1.1). Nevertheless for uniformly bounded transition potential functions, Theorem 2.2 applies to *any* class of marginal distributions.

Last but not least, note that the Weibull marginal distribution with shape parameter strictly less than one is locally bounded on $\mathbb{X} =]0,\infty[$ but does not have compact sub-level sets; for instance, we have

$$\lambda_U(dx) = \frac{1}{2\sqrt{x}}\ e^{-\sqrt{x}}\ 1_{]0,\infty[}(x)\ dx \Longrightarrow U(x) = \sqrt{x} + \log\sqrt{x} + \log 2 \stackrel{x\to 0^+}{\longrightarrow} -\infty.$$

The potential function U above is well defined on the whole state space but does not have compact sub-level sets. Even for bounded transition potential functions Theorem 2.3 cannot be used (but as mentioned above, Theorem 2.2 applies to any marginal models).

3. Contraction of Markov operators

Section 3.1 is dedicated to a brief description of the Hilbert (projective) metric and integral operator norms. Section 3.2 recalls some definitions of contraction coefficients for Markov operators in the context of Hilbert projective metrics, weighted total variation norms and Φ-entropies. These coefficients will be key to the analysis of contraction properties of Sinkhorn semigroups in Section 4.

3.1. Metrics and divergences.

3.1.1. *Hilbert projective metric.* The Hilbert (projective) metric is defined, for any comparable measures $\nu_1, \nu_2 \in \mathcal{M}_1(\mathbb{X})$, by the formula

$$H(\nu_1,\nu_2) := \log\left(\left\|\frac{d\nu_1}{d\nu_2}\right\|\left\|\frac{d\nu_2}{d\nu_1}\right\|\right) \geqslant (\log 3)\ \|\nu_1-\nu_2\|_{\mathsf{tv}}, \tag{3.1}$$

where $\|\cdot\|$ and $\|\cdot\|_{\mathsf{tv}}$ denote the supremum and total variation norms, respectively, and $H(\nu_1,\nu_2) = \infty$ when the measures ν_1, ν_2 are not comparable. The proof of the r.h.s. estimate in (3.1) is provided in [8] (cf. Lemma 1), see also [62]. For a more thorough discussion on Hilbert projective metrics we refer the reader to the recent review and self contained article on Hilbert projective metrics [25].

3.1.2. *Integral operator norms.* Given some functions $\varphi \in \mathcal{B}(\mathbb{X})$ such that $\varphi \geqslant 1/2$, let $\mathcal{M}_\varphi(\mathbb{X}) \subset \mathcal{M}_b(\mathbb{X})$ be the subspace of measures $\nu \in \mathcal{M}_b(\mathbb{X})$ equipped with the operator φ-norm

$$\begin{aligned} \|\!|\nu|\!\|_\varphi &:= \sup\{|\nu(f)|\ :\ \|f\|_\varphi \leqslant 1\} = \sup\{|\nu(f)|\ :\ \mathrm{osc}_\varphi(f) \leqslant 1\} \\ &= |\nu|(\varphi) \quad \text{with the total variation measure} \quad |\nu| := \nu_+ + \nu_-. \end{aligned} \tag{3.2}$$